

Sprint #0 Report

Instructions

Objectives

- Make decisions on the Solitaire software development project.
- Learn unit testing and GUI programming in the language of your choice.

Deliverables and Grading Policy

Read the “CS 449 Homework Overview” document **carefully** and make the key decisions for the software development. Use the following template to complete your report.

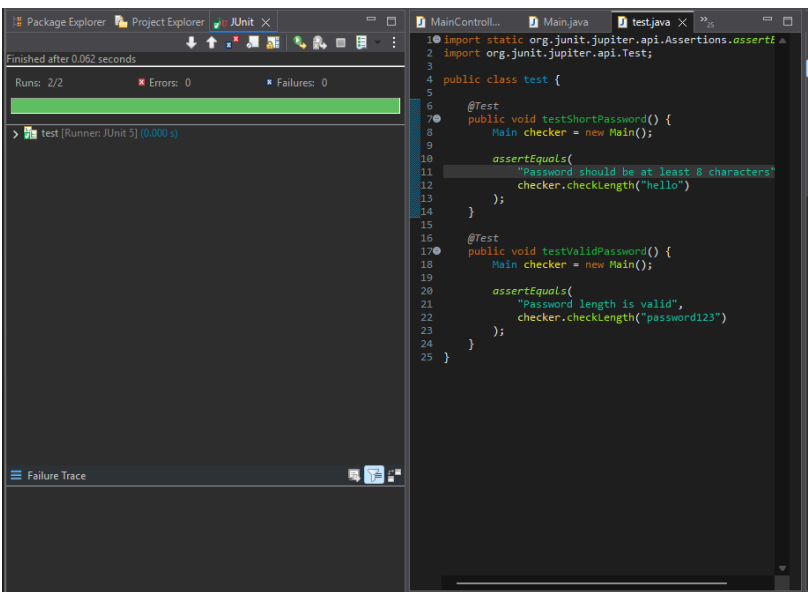
1. Key Decisions of the Solitaire Project (2 points)

Topic	Decision
Object-oriented programming language (if not using OOP, then justify the use of the language)	Java
GUI library (strongly encouraged)	Java swing
IDE (Integrated Development Environment)	Eclipse
xUnit framework (e.g., JUnit for Java)	JUnit
Programming style guide (must read it carefully)	Google Java Style Guide
Project hosting site	Github.com
Other decisions if applicable	

Sample programming style guides:

- Google Java Style Guide: <https://google.github.io/styleguide/javaguide.html>
- Google C++ Style Guide: <https://google.github.io/styleguide/cppguide.html>
- Google Python Style Guide: <https://google.github.io/styleguide/pyguide.html>

2. Unit testing (4 points) -> JUnit5



Main.java:

```
1 public class Main {
2
3     public String checkLength(String password) {
4
5         if (password.length() < 8) {
6             return "Password should be at least 8 characters";
7         } else {
8             return "Password length is valid";
9         }
10    }
11 }
```

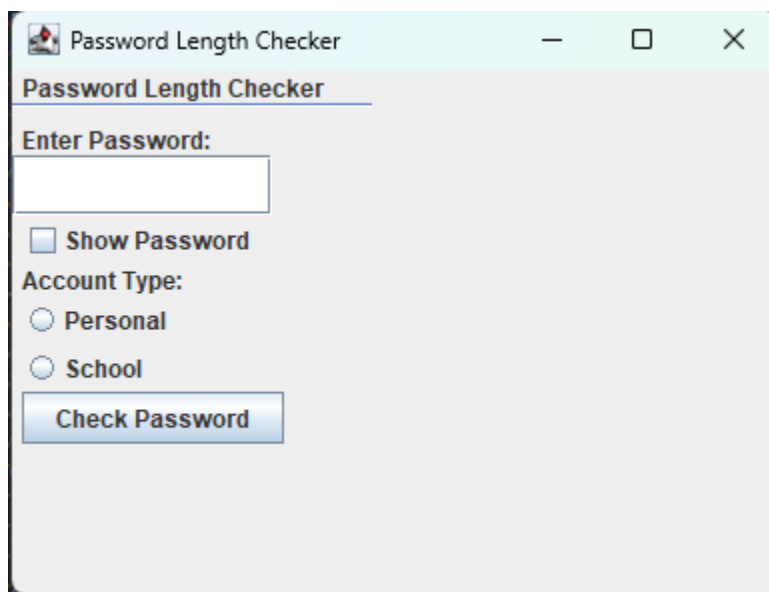
Test.java

```
1 import static org.junit.jupiter.api.Assertions.assertEquals;
2 import org.junit.jupiter.api.Test;
3
4 public class test {
5
6     @Test
7     public void testShortPassword() {
8         Main checker = new Main();
9
10        assertEquals(
11            "Password should be at least 8 characters",
12            checker.checkLength("hello")
13        );
14    }
15
16    @Test
17    public void testValidPassword() {
18        Main checker = new Main();
19
20        assertEquals(
21            "Password length is valid",
22            checker.checkLength("password123")
23        );
24    }
25 }
```

3. GUI programming (4 points)

Write a GUI program in the language you have chosen for your Solitaire project. The GUI of your program must include text, lines, a check box, and radio buttons. In this assignment, any GUI program of your own work is acceptable.

Attach here (1) the screenshot of your program execution and (2) the source code of your program.



```

import java.awt.Dimension;
import javax.swing.*;
public class PasswordGUI {
    public static void main(String[] args) {

        //CREATING THE WINDOW
        JFrame frame = new JFrame("Password Length Checker");
        //SETTING THE SIZE OF THE WINDOW
        frame.setSize(400, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JPanel panel = new JPanel();
        panel.setLayout(new BoxLayout(panel, BoxLayout.Y_AXIS));
        // CREATING THE TITLE TEXT
        JLabel title = new JLabel("Password Length Checker");
        panel.add(title);
        JSeparator line = new JSeparator();
        line.setMaximumSize(new Dimension(350, 10));
        panel.add(line);
        // CREATING THE PASSSSWORD BOX
        JLabel passwordLabel = new JLabel("Enter Password:");
        panel.add(passwordLabel);
        JPasswordField passwordField = new JPasswordField();
        passwordField.setMaximumSize(new Dimension(250, 30));
        panel.add(passwordField);

        JCheckBox showPassword = new JCheckBox("Show Password");
        panel.add(showPassword);
        JLabel accountLabel = new JLabel("Account Type:");
        panel.add(accountLabel);
        JRadioButton personal = new JRadioButton("Personal");
        JRadioButton school = new JRadioButton("School");
        ButtonGroup group = new ButtonGroup();
        group.add(personal);
        group.add(school);
        panel.add(personal);
        panel.add(school);
        // CREATING THE RADIO BUTTON
        JButton checkButton = new JButton("Check Password");
        panel.add(checkButton);
        // Result
        JLabel result = new JLabel("");
        panel.add(result);

        showPassword.addActionListener(e -> {
            if (showPassword.isSelected()) {
                passwordField.setEchoChar((char) 0);
            } else {
                passwordField.setEchoChar('*');
            }
        });
        // Check password button
        checkButton.addActionListener(e -> {
            String password =
                new String(passwordField.getPassword());
            Main checker = new Main();
            result.setText(checker.checkLength(password));
        });
        frame.add(panel);
        frame.setVisible(true);
    }
}

```